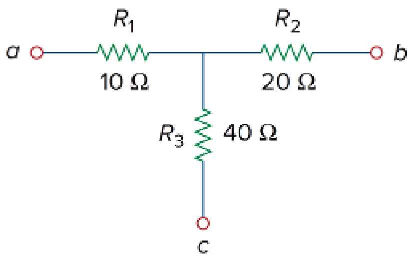
Chapter 2, Problem 14PP

Problem

Transform the wye network in Fig. 2.51 to a delta network.

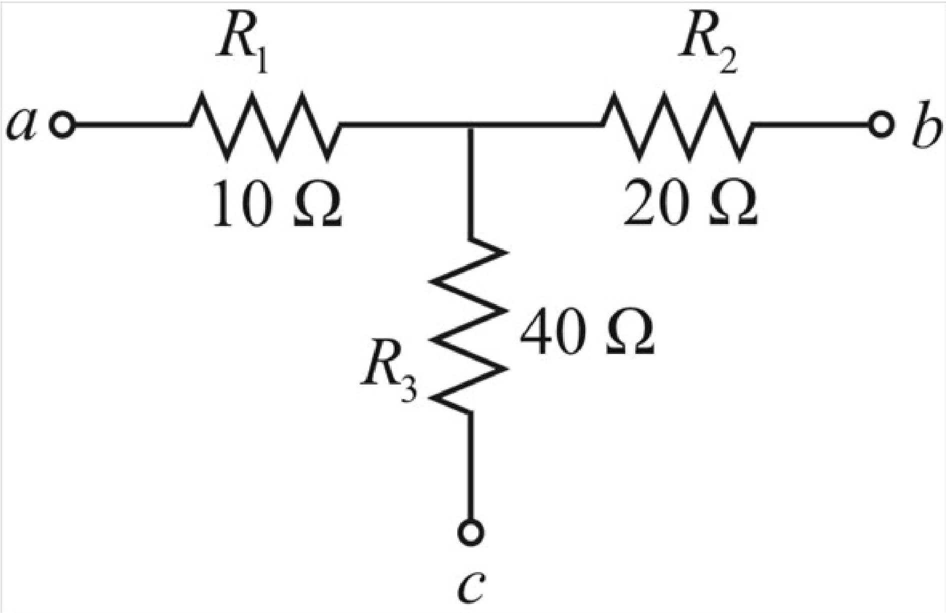
Figure 2.51:



Step-by-step solution

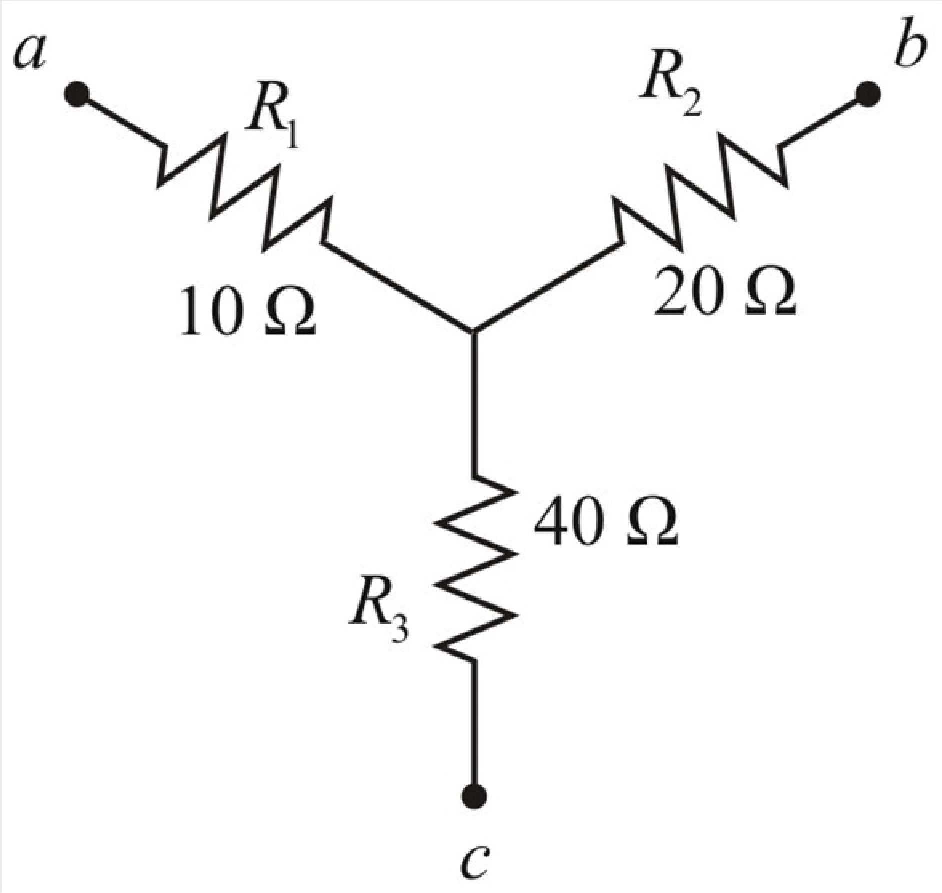
Step 1 of 6

The given wye network is:



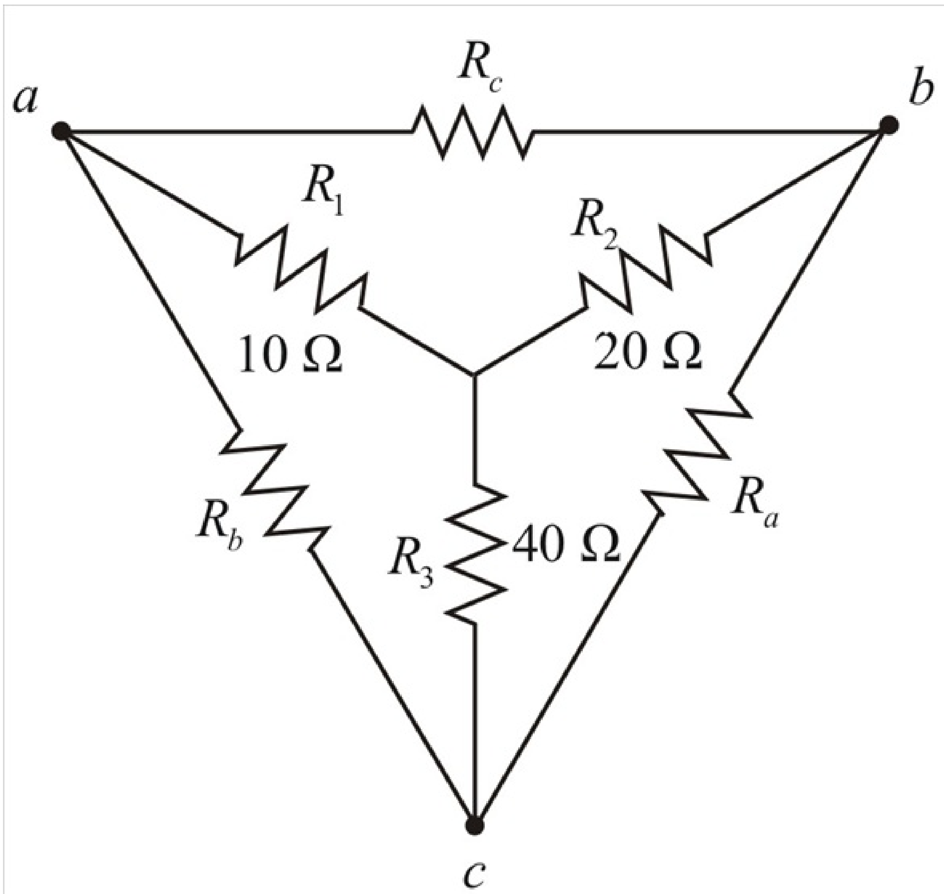
Step 2 of 6

Representing the above network in wye network is



Step 3 of 6

Δ equivalent network is



Step 4 of 6

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

$$R_a = \frac{(10\times 20)+(20\times 40)+(40\times 10)}{10}$$

$$R_a = \frac{200+800+400}{10}$$

$$R_a = \frac{1400}{10}$$

$$\therefore \boxed{R_a = 140\ \Omega}$$

Step 5 of 6

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_b = \frac{(10\times 20)+(20\times 40)+(40\times 10)}{20}$$

$$R_b = \frac{200+800+400}{20}$$

$$R_b = \frac{1400}{20}$$

$$\therefore \boxed{R_b = 70\ \Omega}$$

Step 6 of 6

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$

$$R_c = \frac{(10\times 20)+(20\times 40)+(40\times 10)}{40}$$

$$R_c = \frac{200+800+400}{40}$$

$$R_c = \frac{1400}{40}$$

$$\therefore \boxed{R_c = 35\ \Omega}$$